

OVERVIEW AND OBJECTIVE

I have six years of experience in AI/ML for medical imaging and healthcare, split between research and commercial application. I am seeking a job where I can help researchers doing exciting scientific research, leveraging my skills in HPC, machine learning, and software engineering. I am hard working, dependable, and a quick learner. In addition to my skills in data science and machine learning I also excel at writing, communication, and time management.

COMPUTER SKILLS

- My primary programming language is **Python**.
- Python libraries: **pytorch**, **scikit-learn**, **pandas**, **numpy**, **pyarrow**, **polars**, **matplotlib**, **keras**
- I have a lot of recent experience with **SQL** and extensive past experience with **Fortran**, **Matlab**, **C++**, and **Mathematica**.
- Extensive knowledge of: **git**, **Docker**, **Kubernetes**, **GNU/Linux**, **HPC**, **Slurm**, **bash**, **LaTeX**, **Markdown**, **DICOM**, **OMOP**

WORK EXPERIENCE

Founder and Director

The Metascience Observatory

10/10/2025 — present

Burlington, MA

- Created the world's largest collection of information on replication experiments, spanning all of science, enabling research on how reproducibility varies across fields.
- Created an AI agent equipped with 30+ tools for detecting statistical inconsistencies and data-integrity anomalies in scientific papers.
- experimented with new forms of AI-powered scientific literature review.

Scientist

Radvac

10/10/2025 — present

Waltham, MA

- Research on AI for antiviral discovery

Staff Scientist

National Institutes of Health

10/20/2024 — 10/05/2025

Bethesda, MD

- Developed a transformer model for the prediction of future disease risk using electronic health records (EHRs), as a proof of concept. Currently writing up results.
- Helped plan a data science challenge which would involve using EHR, genetics, wearable, and survey data in the *All of Us* biobank to predict the onset of future disease phenotypes.

Translational Data Scientist

Mass General Brigham

07/21/2021 — 10/7/2024

Somerville, MA

- Took multiple AI/ML systems from research labs and deployed them for testing and validation in the radiology clinic, working closely with researchers, clinicians, engineers, and industry partners.
- Deployed a suite of segmentation algorithms for automated biomarker extraction and ran it on over 250,000 CT scans, creating a resource for future research.
- Helped develop a no-code machine learning framework that allows radiologists to train their own models. Used the system to develop a model for detecting ultrasound images with PHI/PII.
- Helped develop, test, and deploy a model that detects urgent anomalies in head CT, including the use of a conformal prediction method for statistical guarantees on performance. It is currently being tested at MGH for a triage application.
- Developed machine learning models for image classification, segmentation, and anomaly detection.
- Created DICOM and FHIR-formatted AI outputs. Designed and set up DICOM routing rules to route images to AI models.

Staff Scientist (Kelly Government Solutions Contractor)

National Institutes of Health

01/14/2019 — 07/16/2021

Bethesda, MD

- Worked on deep learning-based algorithms that extract biomarkers from CT scans for disease risk prediction, including segmentation and measurement algorithms for plaque, visceral/subcutaneous fat, bone mineral density, muscle, and liver fat.
- Developed a patch-based 3D U-Net model for segmentation of plaque in the aorta and pelvic arteries. The system produced plaque scores that correlated well with manual measurements ($r^2 = 0.94$) on a large external test set.
- Developed a deep learning based system for kidney stone detection that achieved state-of-the-art performance on a challenging set of low-dose CT scans.
- Developed a state-of-the-art model for pancreas segmentation on non-contrast CT using active learning. Ran the model on over 20,000 scans to correlate pancreas features with diabetes diagnosis. The results were published in *Radiology*.
- Constructed a dataset of about 20,000 MRI scans and performed registration of MRI sequences.
- Used the CycleGAN and UNIT image translation models for synthetic data augmentation.
- Performed GPU server installation, maintenance, and backups.
- Co-authored 35 papers published in peer-reviewed journals and conferences. Supervised and mentored a post-baccalaureate fellow and three summer interns.

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Daniel C. Elton, Ph.D.

Data Scientist

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Postdoctoral Researcher

University of Maryland

06/01/2017 — 01/04/2019

College Park, MD

- Applied machine learning techniques to predict the properties of molecules, and used natural language processing techniques to find associations between concepts in the scientific literature. Compared PCA, ICA, and IVA for dimensionality reduction.
- Wrote a review on deep generative modeling for molecular discovery, and co-authored seven peer-reviewed papers total.

Graduate Research Assistant

Stony Brook University

05/07/2012 — 12/15/2016

Stony Brook, NY

- Wrote a Fortran code (*PIMD-F90*) for simulating nuclear quantum effects in liquid water, and a Python package (*spectrumfitter*) for fitting dielectric spectra, including novel techniques. Published four papers in peer-reviewed journals.

EDUCATION

Doctor of Philosophy in Physics, *Stony Brook University*, Stony Brook, NY

Dec. 2016

Bachelor of Science in Physics, *Rensselaer Polytechnic Institute*, Troy, NY

May 2010